

Third-Order Polynomial Approximation of a Nonlinear Irregular Curve

Abstract

This short study demonstrates cubic polynomial regression for approximating an irregular nonlinear function. A synthetic nonlinear curve was sampled, fitted using least-squares polynomial regression, and evaluated using RMSE, MAE, and R^2 .

Method

Original curve: $y = \sin(x) + 0.1x^2 + 0.25\sin(2.5x)$, sampled on $0 \leq x \leq 10$.

Cubic fitted model: $y = -0.011136x^3 + 0.302470x^2 - 1.048169x + 1.448989$

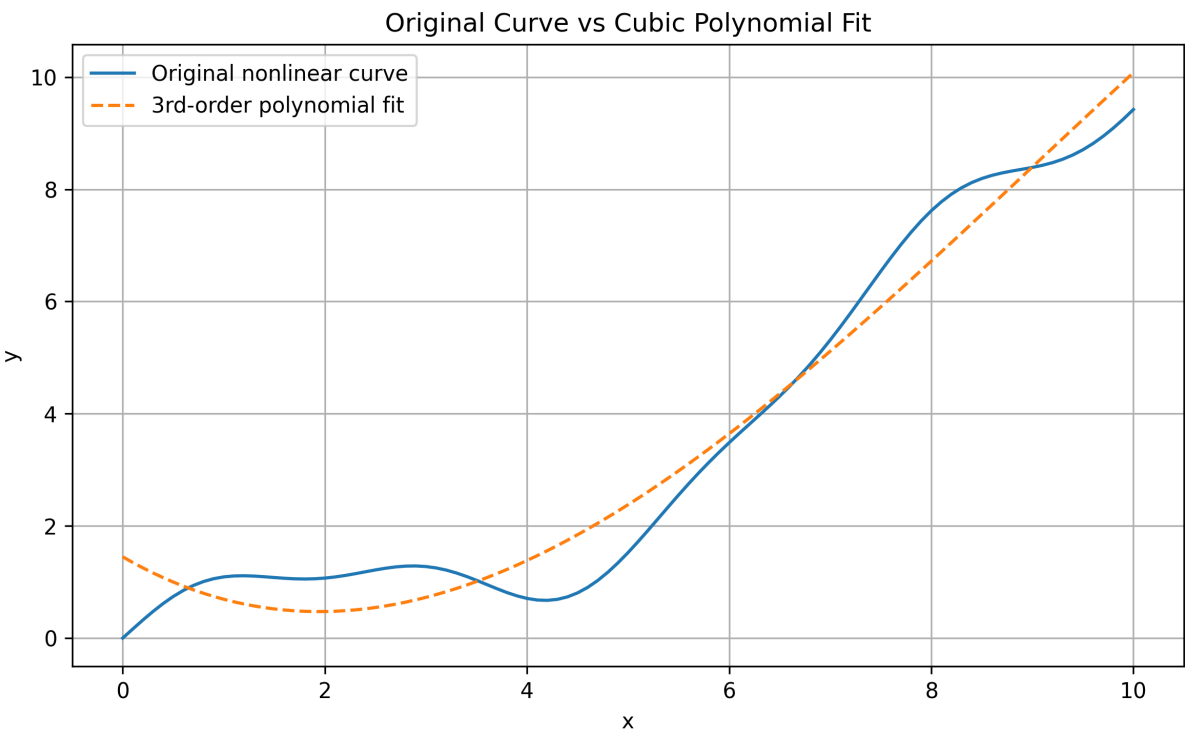


Figure 1. Original nonlinear curve compared with the cubic polynomial fit.

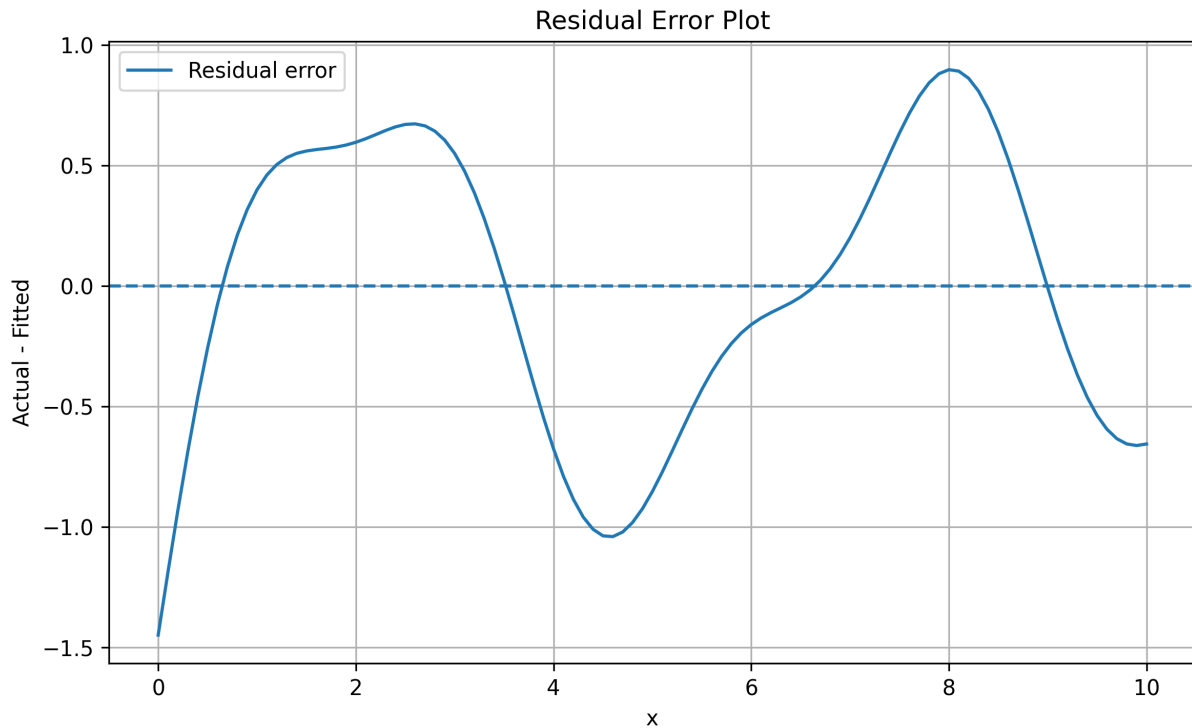


Figure 2. Residual error between original function and fitted polynomial.

Results

RMSE = 0.600195, MAE = 0.519789, $R^2 = 0.962007$.

x	actual_y	fitted_y	residual	absolute_error	squared_error
0.0	0.0	1.449	-1.449	1.449	2.0996
1.0	1.0911	0.6922	0.3989	0.3989	0.1591
2.0	1.0696	0.4734	0.5961	0.5961	0.3554
3.0	1.2756	0.7261	0.5496	0.5496	0.302
4.0	0.7072	1.3832	-0.676	0.676	0.4569
5.0	1.5245	2.3779	-0.8535	0.8535	0.7284
6.0	3.4832	3.6436	-0.1605	0.1605	0.0257
7.0	5.3131	5.1133	0.1997	0.1997	0.0399
8.0	7.6176	6.7203	0.8973	0.8973	0.8051
9.0	8.3903	8.3977	-0.0074	0.0074	0.0001
10.0	9.4229	10.0788	-0.6559	0.6559	0.4302

Conclusion

The third-order polynomial captures the dominant trend of the nonlinear curve while preserving a simple interpretable model. Some local deviations remain because the original function contains oscillatory components that cannot be fully represented by a cubic polynomial.

References

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